

Computer Programming: Lecture 4

Outline

- Conditionals

Conditionals

- Writing programs, we almost always need the ability to check conditions and change the behaviour of the program accordingly.
- Conditionals are used to make decisions in Python. They allow you to run different code depending on the value of a variable or expression.

Conditionals

Simple conditional statement:

if condition:

 # code to run if condition is true

- Python uses indentation to delineate blocks of code
- Most other programming languages use {}

Conditionals

```
name = "John Doe"
```

```
if name == "John Doe":  
    print("Hello, world!")
```

Chained Conditionals

- Elif statements are used to run code if a condition is not true, but another condition is true.

Note: Each condition is checked in order. If the first is false, the next is checked, and so on. If one of them is true, the corresponding branch runs and the statement ends. Even if more than one condition is true, only the first true branch runs.

```
if condition1:  
    # code to run if condition1 is  
    true  
elif condition2:  
    # code to run if condition2 is  
    true
```

Conditionals

```
name = "John Doe"  
if name == "John Doe":  
    print("Hello, John Doe!")  
elif name == "Jane Doe":  
    print("Hello, Jane Doe!")
```

Conditionals

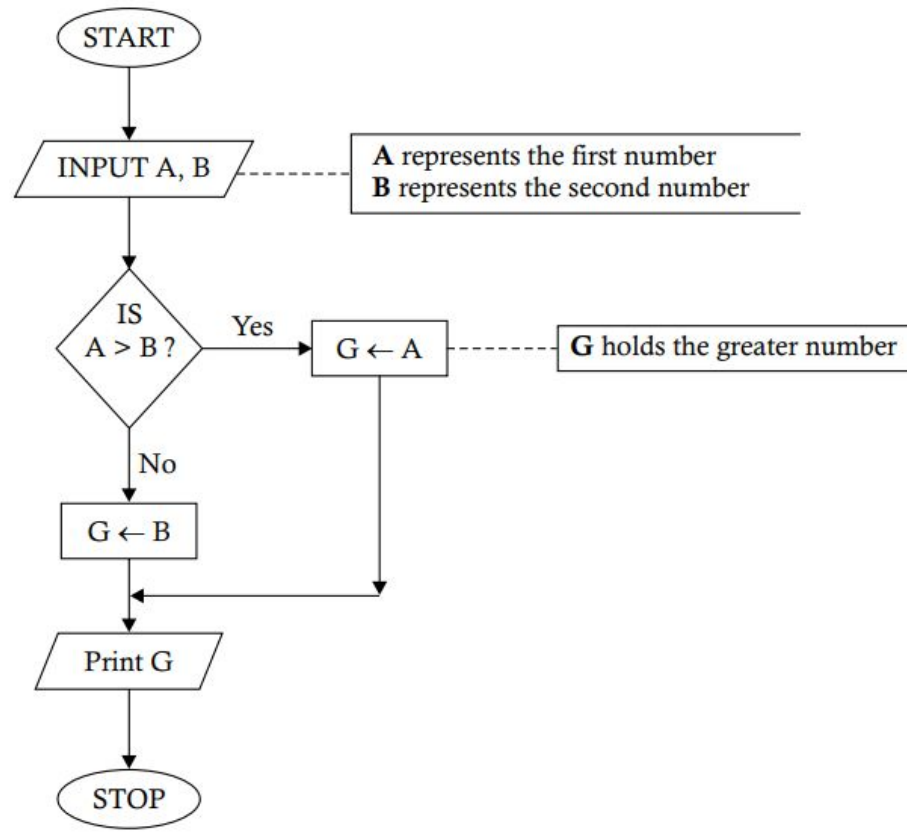
Else statements are used to run code if none of the other conditions are true.

```
if condition1:  
    # code to run if condition1 is true  
elif condition2:  
    # code to run if condition2 is true  
else:  
    # code to run if none of the conditions are true
```


Conditionals

```
name = "John Doe"
if name == "John Doe":
    print("Hello, John Doe!")
elif name == "Jane Doe":
    print("Hello, Jane Doe!")
else:
    print("Hello, stranger!")
```

Example



Demo

Example

Write a function for a computation that accepts two numbers and tells if the first number is a multiple of the second number

Demo

Nested Conditionals

if(<conditional statements>):

 if(<conditional statements>):

 <if block>

 else:

 <else block>

else:

 <else block>

Example

Write a function that checks if a number is divisible by 2 and 3.

Demo

Common Errors

```
if((Value > 0) or (Value <= 10)):  
    print(Value)
```

```
if((Value > 0) and (Value <= 10)):  
    print(Value)
```

Common Errors

```
if((Value < 0) and (Value > 10)):  
    print(Value)
```

Common Errors

```
if((1.11 - 1.10) == (2.11 - 2.10)):  
    print('Equal')
```